

Syed Naveed Mahmood

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[Website](#) | [LinkedIn](#) | [GitHub](#) | [arXiv: 2601.10566](#) | [HyperCite](#)

PROFILE

Computer Science and Engineering graduate from BRAC University (CGPA 3.95/4.00) and Contractual Faculty at Daffodil International University. My research spans trustworthy AI, LLM safety and security, machine unlearning, mechanistic interpretability, secure agentic AI, reproducibility, and explainable AI for health. I have authored an ACL Rolling Review manuscript on representation-aware unlearning and a Scopus-indexed IEEE paper on interpretable EEG sleep staging. I am also interested in extending representation-level methods to vision-language and multimodal models, anomaly detection, and robust adaptation under distribution shift.

RESEARCH INTERESTS

LLM Safety and Security | Machine Unlearning | Mechanistic Interpretability | Representation-Level Auditing | Secure Agentic AI | Trustworthy NLP | Explainable AI for Health | Vision-Language and Multimodal Learning | Anomaly Detection and Robust Adaptation

PROFESSIONAL AND TEACHING EXPERIENCE

Contractual Faculty

Daffodil International University

2026 - Present

Dhaka, Bangladesh

- Teach undergraduate computer science courses using concept-first explanations, practical examples, and structured assessments.
- Prepare lecture materials, evaluate student work, and support students in developing strong software and problem-solving foundations.

Undergraduate Teaching Assistant

BRAC University

Fall 2024 - Fall 2025

CSE220 and CSE251

- Conducted laboratory sessions for Data Structures and Electronic Devices and Circuits for more than 100 students.
- Evaluated assignments, quizzes, and examinations and provided detailed technical feedback and student guidance.

EDUCATION AND TEST SCORES

B.Sc. in Computer Science and Engineering

BRAC University

Graduated 2026

CGPA: 3.95/4.00

Dean's List and VC's List for 10 consecutive semesters | 75% Merit-Based Tuition Waiver

IELTS Academic: 8.0 overall | Listening: 8.5 | Reading: 8.0 | Writing: 7.0 | Speaking: 7.5

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher

BRAC University

2025 - 2026

Trustworthy AI and Machine Unlearning

Representation-Aware Unlearning via Activation Signatures: From Suppression to Entity-Signature Erasure

Submitted to ACL Rolling Review, May 2026 Cycle; arXiv: 2601.10566

- Developed ERUF, a representation-aware framework for evaluating and reducing targeted knowledge while preserving general utility.
- Designed hidden-state diagnostics and adversarial recovery tests to distinguish internal attenuation from output-level suppression or obfuscation.
- Evaluated language models from 3B to 32B parameters across multiple model families and machine-unlearning benchmarks.

Researcher

BlackboxNLP Reproducibility Task

2026 - Present

Paper in Progress

- Reproducing and stress-testing a proposed mechanistic account of transformer attention sinks using controlled causal interventions.
- Built multi-seed and cross-checkpoint evaluation pipelines to examine implementation sensitivity and the reliability of mechanistic conclusions.

ADDITIONAL RESEARCH EXPERIENCE

Independent Researcher

Secure and Interpretable LLM Agents

- Investigating representation-level diagnostics and lightweight defenses for indirect prompt injection and goal manipulation in tool-using agents.
- Developing evaluation procedures that test whether unsafe information remains active internally even when final actions appear safe.

2026 - Present

Ongoing Research

Deep Learning Researcher

Interpretable and Transferable EEG Sleep Staging

- Studying interpretable sleep-stage classification and representation transfer across datasets, recording settings, and subject populations.
- Evaluating masked representation learning and morphology-aware modeling for robust EEG classification.

2025 - Present

Ongoing Research

Deep Learning Researcher

BRAC University

- Developed an attention-based CNN with branch-aware Dual-Mode GradCAM for EEG sleep staging and published the work at IEEE ICITEE 2025.

2024 - 2025

Explainable Biomedical AI

PUBLICATIONS AND MANUSCRIPTS

Mahmood, S. N. (2026). Representation-Aware Unlearning via Activation Signatures: From Suppression to Entity-Signature Erasure. Submitted to ACL Rolling Review, May 2026 Cycle. [arXiv:2601.10566](https://arxiv.org/abs/2601.10566).

Mahmood, S. N., Ferdous, T., and Zereen, A. N. (2025). GradG-AttnSleep: Dual-Mode GradCAM for Interpretable Sleep Staging with an Attention-Based CNN. IEEE ICITEE 2025. [doi:10.1109/ICITEE66631.2025.11338388](https://doi.org/10.1109/ICITEE66631.2025.11338388). Scopus indexed.

Mahmood, S. N. et al. (2026). Same Sink, Different Plumbing: Reproducing and Stress-Testing the GPT-2 Attention-Sink Circuit. BlackboxNLP Reproducibility Task. Paper in progress.

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, PEFT/LoRA, NNSight, GradCAM, Pandas, NumPy

Research Methods: Activation hooks, hidden-state diagnostics, causal intervention, adversarial evaluation, robustness benchmarking, knowledge distillation, utility-preservation analysis, experiment harness design

Vision and Multimodal AI: CNNs, transformer attention, GradCAM, vision-language model evaluation, multimodal anomaly detection, domain generalization, test-time adaptation

Programming: Python, C, C++, Java, JavaScript, TypeScript, PHP, SQL, Assembly

Systems and Web: Linux, Git, Conda, Slurm, Jupyter, LaTeX, React, Node.js, Express.js, MongoDB, MySQL, Vercel

SELECTED PROJECTS

HyperCite: AI Research Summarization and Collaboration Hub | TypeScript, Vite, Vercel

AI-assisted research workspace for paper exploration, summarization, and collaboration. CSE471 Best Project Award, Fall 2025. [Live project](#)

PaperDebate Agent | Python, Local LLM Workflows

Local-first research companion for structured paper analysis and multi-perspective scholarly discussion. [GitHub](#)

Dual-MPU Kyphotic-Lordotic Posture Monitor | IoT, Embedded Systems

Wearable posture-monitoring system using dual inertial sensors, haptic feedback, and real-time analytics.

AWARDS AND HONORS

Dean's List and VC's List Honoree, 10 consecutive semesters | 75% Merit-Based Tuition Waiver | Second Place, Undergraduate Thesis Poster Presentation, Summer 2025 | CSE471 Best Project Award, Fall 2025 | Scopus-Indexed IEEE Publication, ICITEE 2025

CERTIFICATIONS

DeepLearning.AI TensorFlow Developer Professional Certificate | Coursera | September 2024

Human Research: Data or Specimens Only Research, Refresher | CITI Program | August 2024